# Net Price, Student Composition, and Long-Term Earnings Across U.S. Colleges

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**Abstract**

This report examines how the cost of attendance and socioeconomic composition of a student body relate to the long-term earnings of college graduates, using data from the College Scorecard Institution-Level Dataset. Each observational unit is a postsecondary institution, and the response variable is median earnings of former students 10 years after entry. We find that while net price is positively associated with earnings, its effect is modest once student characteristics are accounted for. The proportion of students receiving Pell Grants is negatively associated with earnings even after controlling for institutional characteristics. Average SAT score emerges as the strongest predictor, suggesting that student academic preparation drives long-term earnings outcomes more than institutional cost alone.

# 1 Introduction

How does the cost of attending a college and the socioeconomic composition of its student body relate to the long-term earnings of its graduates? As tuition costs continue to rise, students and families face increasingly consequential decisions about where to enroll. A common assumption is that more expensive institutions produce better financial outcomes, but it is unclear whether this reflects the value added by the institution itself or simply the fact that more selective schools enroll stronger students to begin with. We are using data from the College Scorecard Institution-Level Dataset to examine how average net price and the percentage of students receiving Pell Grants are associated with median earnings of former students 10 years after entry, after accounting for institutional size and type. We investigate two questions:

1. Is the average net price associated with median earnings 10 years after entry, and does this relationship differ between public and private institutions?
2. Is the proportion of Pell Grant recipients associated with median earnings, after controlling for net price and institutional characteristics?

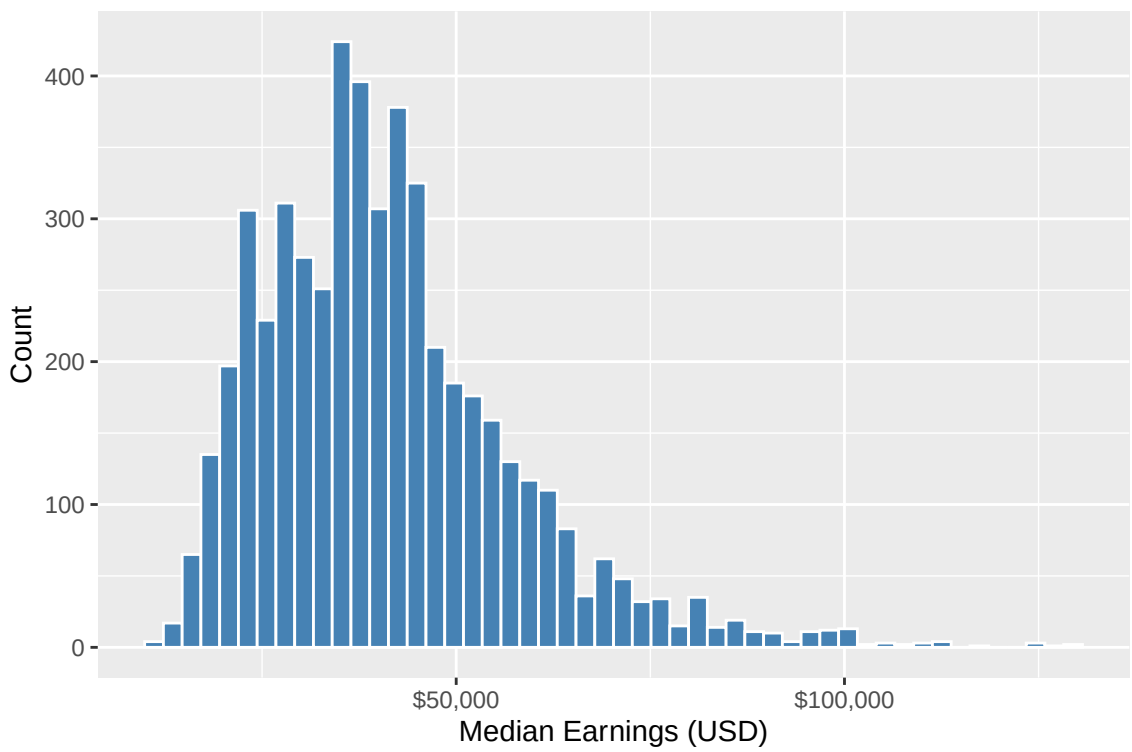


*Figure 1: Distribution of median earnings 10 years after college entry across institutions. The right-skewed shape motivates the use of a log transformation in all regression models.*

## 1.1 Data Description

The data used in this analysis come from the College Scorecard Institution-Level Dataset, a publicly available dataset published by the U.S. Department of Education. Each observational unit is a postsecondary institution. After removing observations with missing values

on the variables used in our final model, 961 institutions remain for the primary analysis, though simpler models that exclude SAT score retain up to 4,455 institutions. The response variable is median earnings of former students 10 years after entry, measured in dollars. Earnings range from roughly $11,000 to $129,000 across institutions, with a median of approximately $38,700. Because the distribution of earnings is right-skewed, we apply a log transformation for all regression models. The primary predictors of interest are average net price and the proportion of students receiving Pell Grants. Net price represents the average cost of attendance after financial aid, measured in dollars, and ranges from near $0 to approximately $108,000 across institutions. The proportion of Pell Grant recipients ranges from 0 to 1 and serves as a measure of the socioeconomic composition of the student body, with higher values indicating a greater share of low-income students. We also include average SAT score as a proxy for student academic preparation, undergraduate enrollment as a measure of institutional size, and institution type as a categorical variable distinguishing public, private nonprofit, and private for-profit schools.

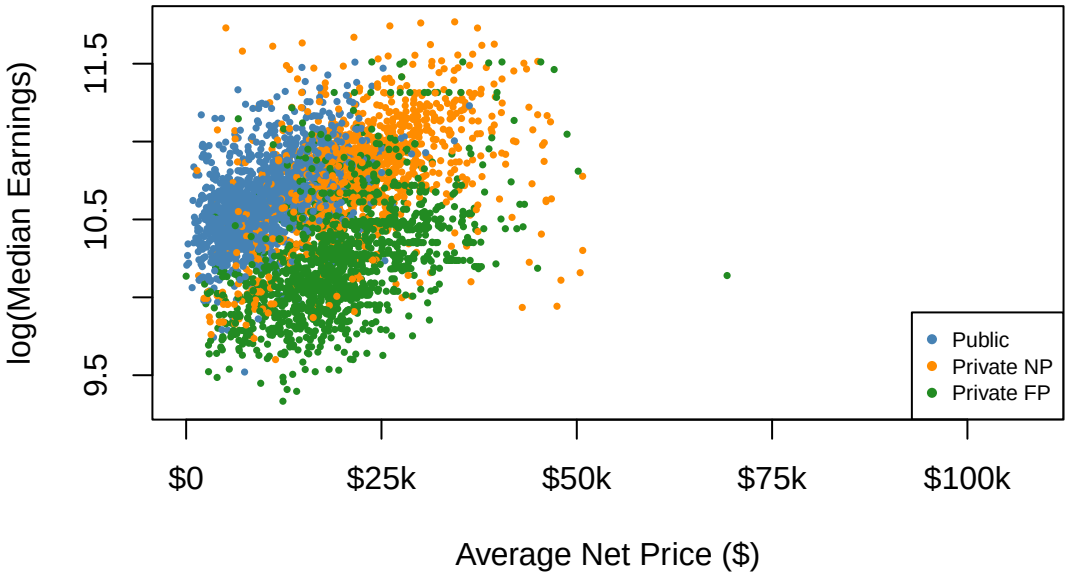


*Figure 2: Average net price versus log median earnings 10 years after entry, color-coded by institution type. A modest positive relationship is visible across all three institution types.*

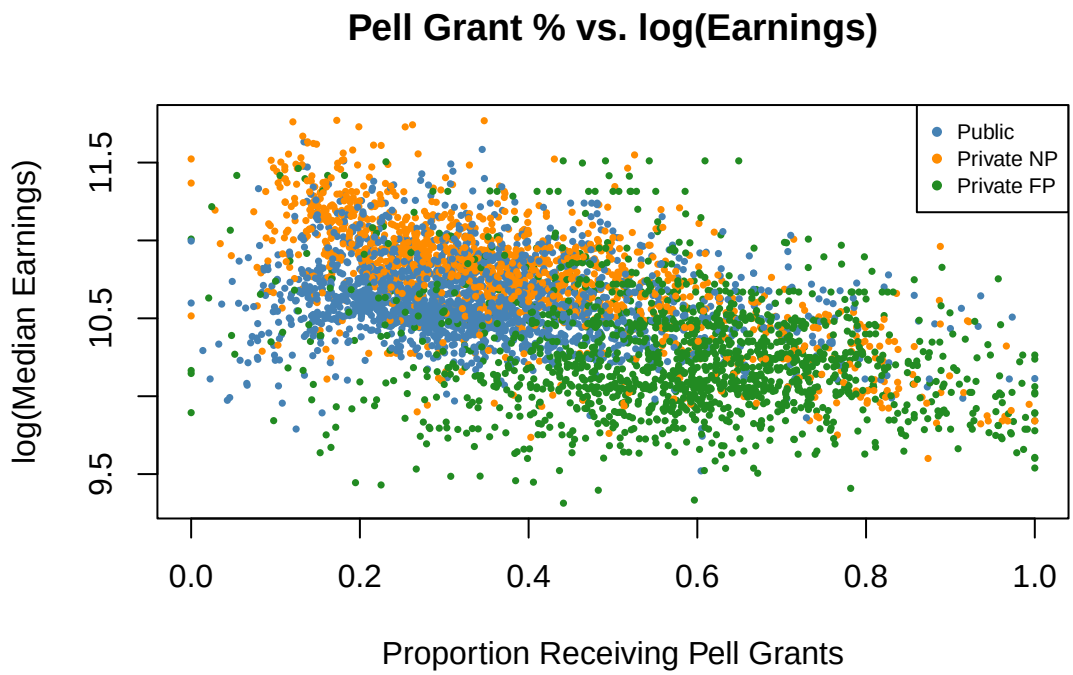


*Figure 3: Proportion of students receiving Pell Grants versus log median earnings 10 years after entry, color-coded by institution type. A clear negative relationship is visible across all three institution types, suggesting that institutions serving more low-income students tend to have lower graduate earnings outcomes.*

## 2 Model Selection and Validation

To analyze the relationship between institutional characteristics and long-term earnings, we fit a multiple linear regression model with log-transformed median earnings as the response variable. The model is given by

$$\begin{aligned}\log(\text{Earnings}_i) = \beta_0 &+ \beta_1 \cdot \text{NetPrice}_i \\ &+ \beta_2 \cdot \text{PellShare}_i \\ &+ \beta_3 \cdot \text{SAT}_i \\ &+ \beta_4 \cdot \text{Enrollment}_i \\ &+ \beta_5 \cdot \text{I}(\text{Control}_i = \text{Private Nonprofit}) \\ &+ \beta_6 \cdot \text{I}(\text{Control}_i = \text{Private For-Profit}) + \epsilon_i,\end{aligned} \tag{1}$$

where $\epsilon_i \sim N(0, \sigma^2)$.

The response variable is the logarithm of median earnings 10 years after entry. We apply a log transformation because exploratory data analysis showed that the distribution of earnings is right-skewed, and the transformation improves linearity and stabilizes variance.

The predictor variables were selected based on both theoretical relevance and exploratory analysis. Net price represents the cost of attending an institution and is central to our primary research question. The proportion of students receiving Pell Grants is included to

capture socioeconomic composition, while average SAT score serves as a proxy for student academic preparation. Undergraduate enrollment is included as a measure of institutional size. We also include institution type as a categorical variable to account for structural differences between public, private nonprofit, and private for-profit institutions.

We initially considered more complex models that included interaction terms between net price and institution type, as well as additional variables such as admission rate. However, these terms were either not statistically significant or did not meaningfully improve model fit and AIC did not improve. As a result, we selected a more parsimonious model that balances interpretability and predictive performance.

To assess predictive performance and justify our model selection, we conducted a cross-validation study comparing three candidate models: a basic model including only net price and Pell Grant share, a second model adding institution type, and our final model which additionally includes average SAT score and undergraduate enrollment. The final model achieved a mean cross-validated $R^2$ of 0.61 and a root mean squared error of 0.152, outperforming both the basic model ($R^2 = 0.44$, RMSE = 0.181) and the model with institution type ($R^2 = 0.43$, RMSE = 0.184). Adding institution type alone provided little improvement over the basic model, reinforcing the conclusion that differences between public and private institutions are largely explained by student characteristics. The consistency of the final model's $R^2$ across folds suggests the model generalizes well rather than overfitting to the training data.

## 3 Analyses, Results, and Interpretation

We fit the multiple linear regression model described in Section 2 using ordinary least squares. The response variable is the logarithm of median earnings 10 years after entry, and the predictors include net price, Pell Grant share, average SAT score, undergraduate enrollment, and institution type. Table 1 presents the estimated coefficients for the final model.

Because the response variable is log-transformed, the estimated coefficients can be interpreted as approximate percentage changes in earnings associated with a one-unit increase in each predictor, holding all other variables constant.

| | Estimate | Std. Error | t value | Pr(>\|t\|) |
|---|---|---|---|---|
| (Intercept) | 9.715 | 0.070 | 138.01 | $< 2 \times 10^{-16}$ |
| NET_PRICE | $5.69 \times 10^{-6}$ | $9.08 \times 10^{-7}$ | 6.27 | $5.53 \times 10^{-10}$ |
| PCTPELL | -0.264 | 0.052 | -5.06 | $5.07 \times 10^{-7}$ |
| SAT_AVG | 0.00199 | 0.00010 | 19.34 | $< 2 \times 10^{-16}$ |
| UGDS | $1.82 \times 10^{-6}$ | $7.50 \times 10^{-7}$ | 2.43 | 0.0153 |
| factor(CONTROL)2 | -0.0089 | 0.0144 | -0.62 | 0.537 |
| factor(CONTROL)3 | 0.0302 | 0.0897 | 0.34 | 0.736 |

*Table 1: Regression summary output for the final model predicting log median earnings 10 years after entry.*

The fitted model explains a substantial portion of the variability in earnings, with an adjusted $R^2$ of approximately 0.61. This indicates that the included predictors collectively provide a strong explanation of differences in long-term earnings across institutions. The

residual standard error is approximately 0.15, indicating relatively low unexplained variability in log earnings.

The estimated coefficient for average SAT score is positive and highly statistically significant ($p < 2 \times 10^{-16}$). Specifically, a one-point increase in average SAT score is associated with an approximate increase of 0.002 in log earnings, holding other variables constant. This suggests that student academic preparation is a major driver of long-term earnings outcomes.

The proportion of students receiving Pell Grants is negatively associated with earnings ($p < 0.001$). The estimated coefficient indicates that institutions with a higher share of low-income students tend to have lower median earnings, even after controlling for other institutional characteristics. This highlights the importance of socioeconomic composition in explaining differences in earnings outcomes.

Net price remains positively associated with earnings ($p < 0.001$), but the magnitude of this effect is relatively small compared to other predictors. This suggests that while higher-cost institutions are associated with slightly higher earnings, much of this relationship is explained by differences in student characteristics rather than cost alone. This highlights the importance of accounting for confounding variables, as the apparent effect of cost is largely driven by differences in student populations across institutions.

Undergraduate enrollment is also positively associated with earnings ($p < 0.05$), indicating that larger institutions are associated with slightly higher earnings outcomes, though the effect is modest in size.

In contrast, institution type is not a statistically significant predictor after controlling for the other variables in the model. This suggests that differences between public, private nonprofit, and for-profit institutions are largely explained by factors such as student quality and socioeconomic composition, rather than institutional type itself.

Overall, these results indicate that student characteristics, particularly academic preparation and socioeconomic background, play a central role in explaining long-term earnings outcomes. While institutional cost is positively associated with earnings, its effect is relatively small once these other factors are taken into account.

# 4 Conclusions

This analysis examined how the cost of attendance and socioeconomic composition of a student body relate to the long-term earnings of graduates, using data from the College Scorecard Institution-Level Dataset. Our results provide partial support for the common assumption that higher-cost institutions produce better earnings outcomes. Net price remains positively associated with earnings, but the magnitude of this effect is relatively small compared to student characteristics. Furthermore, this relationship does not meaningfully differ between public and private institutions, suggesting that the type of institution matters less than the characteristics of the students it enrolls.

The proportion of students receiving Pell Grants is negatively associated with earnings even after controlling for net price, institutional size, and institution type. This indicates that socioeconomic composition plays an independent role in explaining earnings differences across institutions, likely reflecting barriers that low-income students face beyond the college experience itself.

Taken together, these findings suggest that student characteristics, particularly academic preparation as measured by average SAT score, are the dominant driver of long-term earnings outcomes. Institutional cost appears to matter, but much of its apparent effect may reflect the kinds of students that higher-cost institutions attract rather than any direct value added by the institution.

There are several limitations worth noting. First, because this is an observational study, we cannot draw causal conclusions. Institutions with higher earnings outcomes may differ from lower-earning institutions in ways not captured by our model. Second, median earnings measured 10 years after entry may not fully capture long-term career trajectories, particularly for graduates in fields such as medicine or law where earnings tend to peak later in a career. Third, including average SAT score in the final model reduced the available sample size, as many institutions do not report SAT data. This means our estimates are based primarily on more traditional four-year institutions and may not generalize to the full range of colleges in the dataset. Finally, even with an adjusted $R^2$ of 0.61, a substantial portion of the variation in earnings remains unexplained. Factors such as the geographic location of an institution, the mix of academic programs it offers, and institutional endowment are likely important drivers of earnings outcomes that were not available in our model.

# APPENDIX

**Use of Generative AI for Feedback**

**Prompt:** We used ChatGPT to review our regression analysis, help refine model selection, and improve clarity in our written interpretations and overall report structure.

**Primary Recommendations:**

- Simplify the final model by removing unnecessary variables and interaction terms
- Focus on clearly interpreting the most important predictors (SAT score, Pell share, and net price)
- Improve the flow and readability of the report by adding transitions and clearer explanations
- Avoid overloading the report with unnecessary diagnostic plots and instead emphasize key findings

**Recommendations Not Adopted:** Some suggestions involved adding additional modeling techniques such as LASSO regression or further transformations of predictors. We chose not to include these methods in the final report in order to maintain interpretability and focus on the primary research questions.

**Incorporated Changes:** We incorporated suggestions related to model simplification, interpretation of coefficients, and improving the clarity and structure of the report. In particular, we refined our final model, strengthened the explanation of key results, and improved transitions between sections to create a more cohesive narrative.